InkOrPixel: Medium Classification of Artwork Using Convolutional Neural Networks

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***Abstract*—This paper presents InkOrPixel, a binary image classification system designed to distinguish between traditional artwork (hand-drawn on physical media) and digital artwork (created with illustration software). Unlike conventional image classification tasks that focus on identifying semantic content, this work addresses the problem of inferring the creation medium from low-level visual characteristics such as texture, stroke regularity, noise patterns, and gradient smoothness. Two convolutional neural network (CNN) approaches are investigated: a baseline CNN trained from scratch and a transfer learning approach using a pretrained ResNet-18 model with selective fine-tuning. The system is built on a custom dataset of original artwork authored by the researcher, with data augmentation applied to address the limited dataset size. Experimental results demonstrate that the transfer learning approach achieves higher confidence on unambiguous examples, while both models face challenges with stylistically ambiguous inputs. The findings suggest that CNNs can learn medium-specific representations even from small, specialized datasets when combined with appropriate regularization and augmentation strategies. The complete source code is publicly available at https://github.com/aditiputrevu/InkOrPixel.**

***Index Terms*—convolutional neural networks, image classification, transfer learning, digital artwork, traditional artwork, medium classification, ResNet, texture analysis**

I. Introduction

Image classification has been one of the most extensively studied problems in computer vision and artificial intelligence. The majority of prior work focuses on recognizing the semantic content of images, such as identifying objects, scenes, or activities. However, a less explored but equally interesting problem is classifying the creation medium of an image, particularly in the domain of visual art. Determining whether an artwork was produced using traditional physical media (e.g., pen, ink, or paint on paper) or created digitally using illustration software presents a unique challenge because the distinction often lies in subtle textural and structural differences rather than in the depicted subject matter.

This paper introduces InkOrPixel, a system that frames medium classification as a binary image classification task. Given an input image of artwork, the model predicts whether the piece was created traditionally or digitally. The motivation for this work stems from several practical considerations. Automated medium classification can assist in digital art curation, provenance verification, art authentication workflows, and metadata tagging for large artwork repositories. Furthermore, the problem provides an instructive case study in representation learning, as the model must learn to attend to low-level features such as paper grain, stroke irregularity, color gradient smoothness, and edge sharpness rather than high-level semantic content.

The key contributions of this work are as follows. First, a complete end-to-end classification pipeline is developed, encompassing dataset construction, preprocessing, model training, evaluation, and inference. Second, two modeling approaches are compared: a baseline CNN trained from scratch and a transfer learning approach using a pretrained ResNet-18 architecture. Third, the challenges of training deep learning models on small, domain-specific datasets are examined, with particular attention to overfitting mitigation through data augmentation, dropout regularization, and learning rate scheduling. Fourth, the system is evaluated using accuracy metrics and confidence analysis, with qualitative discussion of failure cases.

The remainder of this paper is organized as follows. Section II reviews related work in image classification and transfer learning. Section III describes the proposed methodology, including dataset construction, model architectures, and training procedures. Section IV presents experimental results and analysis. Section V discusses limitations and future directions. Section VII concludes the paper.

II. Related Work

A. Convolutional Neural Networks for Image Classification

Convolutional neural networks have become the dominant approach for image classification since the landmark work of Krizhevsky et al. [1], who demonstrated that a deep CNN (AlexNet) could achieve substantially lower error rates on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) compared to methods based on handcrafted features. Their network achieved a top-5 error rate of 15.3%, compared to 26.2% for the second-place entry, establishing the

effectiveness of learned hierarchical feature representations for visual recognition tasks.

Several architectural innovations followed. Simonyan and Zisserman [2] showed that increasing network depth with small convolutional filters (VGGNet) improved performance. Szegedy et al. [3] introduced the Inception architecture with parallel convolutional pathways operating at multiple scales. He et al. [4] proposed residual networks (ResNet), which introduced skip connections to address the vanishing gradient problem in very deep networks, enabling architectures with over 100 layers to be trained effectively.

Key techniques that have become standard in CNN-based classification include the Rectified Linear Unit (ReLU) activation function, which addresses the vanishing gradient problem associated with saturating nonlinearities; dropout regularization, which prevents co-adaptation of neurons by randomly disabling units during training; and batch normalization, which stabilizes training by normalizing intermediate activations.

B. Transfer Learning in Computer Vision

Transfer learning has proven highly effective for tasks with limited labeled data. The core idea is to leverage feature representations learned from a large source dataset (typically ImageNet) and adapt them to a smaller target domain. Yosinski et al. [5] investigated the transferability of features across CNN layers and found that early layers learn general features (edges, textures, color patterns) that transfer well across tasks, while later layers become increasingly task-specific.

Common transfer learning strategies include feature extraction, where the pretrained network serves as a fixed feature extractor with only the classification head retrained, and fine-tuning, where some or all pretrained layers are updated with a reduced learning rate. Kornblith et al. [6] showed that models pretrained on ImageNet transfer well to a wide range of downstream tasks, with performance generally correlating with ImageNet accuracy.

C. Texture and Style Classification

The problem addressed in this paper is related to texture classification and artistic style recognition. Gatys et al. [7] demonstrated that CNNs capture both content and style information in their intermediate representations, with style being encoded in the correlations between feature maps (Gram matrices). Karayev et al. [8] applied deep learning to artistic style recognition, classifying images by visual style categories such as impressionism or minimalism.

Medium classification differs from style classification in that it focuses on physical properties of the creation process rather than artistic stylistic choices. The visual cues relevant to medium classification, such as paper texture, scanner artifacts, stroke pressure variation, and pixel-level smoothness, are primarily low-level features that CNNs are well-suited to capture in their early convolutional layers.

III. Methodology

A. Problem Formulation

The medium classification task is formulated as a supervised binary classification problem. Let $\mathcal{X}$ denote the space of RGB images and $\mathcal{Y} = \{0, 1\}$ denote the label space, where 0 corresponds to "digital" and 1 corresponds to "traditional." Given a training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, the objective is to learn a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ parameterized by θ that minimizes the empirical risk:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell(f_\theta(x_i),\, y_i) + \lambda \|\theta\|_2^2 \tag{1}$$

where ℓ denotes the cross-entropy loss function and λ is a weight decay coefficient providing L_2 regularization.

B. Dataset Construction

The dataset was constructed from original artwork created by the author to ensure full ownership and eliminate copyright concerns. Images were divided into two classes:

- **Traditional:** Hand-drawn or ink-based artwork on paper, captured via scanning or photography. These images typically exhibit visible paper texture, organic stroke variation, and minor artifacts from the digitization process (e.g., uneven lighting, slight blur).
- **Digital:** Illustrations created using digital tools (e.g., Procreate, Photoshop). These images are characterized by smoother gradients, more uniform backgrounds, cleaner edge transitions, and the absence of physical media artifacts.

A dedicated preprocessing script handles the partitioning of the raw dataset into training, validation, and test sets using a stratified split. With a validation split ratio of 0.2 and a test split ratio of 0.2, the data is divided such that approximately 60% is used for training, 20% for validation, and 20% for testing. A fixed random seed ensures reproducibility across runs. A minimum of six images per class is enforced to ensure all three splits contain at least one sample.

C. Data Augmentation

Given the limited size of the custom dataset, data augmentation is applied during training to artificially increase variability and reduce overfitting. The augmentation pipeline is applied exclusively to training data and includes the following transformations:

- **Random resized crop:** Images are first resized to 256×256 pixels, then randomly cropped to 224×224 with a scale factor uniformly sampled from $[0.8, 1.0]$.
- **Random horizontal flip:** Each image has a 50% probability of being horizontally mirrored.
- **Random rotation:** Images are randomly rotated by up to ± 15 degrees.
- **Color jitter:** Random perturbations are applied to brightness (± 0.2), contrast (± 0.2), saturation (± 0.2), and hue (± 0.05).

- **Normalization:** All images are normalized using the ImageNet channel-wise mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]).

Validation and test images undergo only resizing to 224×224 and normalization, without any stochastic augmentation, to ensure deterministic evaluation.

D. Model Architectures

1) Baseline CNN: The first approach employs a simple convolutional neural network built from scratch, inspired by architectures commonly used for small-scale image classification benchmarks such as CIFAR-10. This baseline serves as a reference point for evaluating the benefit of transfer learning. The architecture follows a standard pattern of alternating convolutional and pooling layers followed by fully connected layers with ReLU activations and dropout regularization.

2) Transfer Learning with ResNet-18: The primary model uses transfer learning with ResNet-18 [4], a deep residual network pretrained on the ImageNet dataset. ResNet-18 consists of 18 layers organized into four residual blocks, where each block contains skip (shortcut) connections that enable gradient flow through the network and facilitate the training of deeper architectures.

The transfer learning strategy employs selective fine-tuning. All layers up to and including the third residual block (`layer3`) are frozen, preserving the general low-level and mid-level feature representations learned from ImageNet. The fourth residual block (`layer4`) is unfrozen and fine-tuned to adapt higher-level features to the artwork domain. The original fully connected classification head (designed for 1,000 ImageNet classes) is replaced with a new head consisting of a dropout layer (rate $= 0.4$) followed by a linear layer mapping to the two output classes.

This selective fine-tuning approach balances the retention of general visual features with adaptation to the specific characteristics of the artwork classification task. Freezing early layers is particularly advantageous when the target dataset is small, as it reduces the number of trainable parameters and mitigates overfitting.

E. Training Procedure

Models are trained using the Adam optimizer [9] with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-4}. The loss function is cross-entropy loss, appropriate for the binary classification setting. A batch size of 8 is used to accommodate the small dataset size.

A learning rate scheduler (`ReduceLROnPlateau`) monitors validation accuracy and reduces the learning rate by a factor of 0.5 if no improvement is observed for 2 consecutive epochs. Early stopping is employed with a patience of 5 epochs, terminating training if validation accuracy does not improve for 5 consecutive epochs. The model checkpoint with the highest validation accuracy is saved and used for final evaluation.

Training is conducted on the available hardware, with automatic device selection between CUDA-enabled GPU and CPU. A fixed random seed of 42 is used throughout for reproducibility.

F. Evaluation Metrics

Model performance is evaluated using classification accuracy on the held-out test set:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \times 100\% \quad (2)$$

Additionally, prediction confidence is analyzed by examining the softmax probability associated with the predicted class. This provides insight into the model's certainty and helps identify ambiguous cases where the distinction between traditional and digital artwork is unclear.

IV. Experiments and Results

A. Experimental Setup

All experiments were conducted using Python 3.x with PyTorch as the deep learning framework. The project codebase follows a modular architecture with separate components for configuration (`config.py`), dataset handling (`dataset.py`), model definition (`model.py`), training (`train.py`), evaluation (`evaluate.py`), and inference (`predict.py`). The complete source code is publicly available at https://github.com/aditiputrevu/InkOrPixel.

Table I summarizes the hyperparameters used in the experiments.

TABLE I
Training Hyperparameters

Parameter	Value
Input image size	224×224
Batch size	8
Number of epochs	20 (max)
Learning rate	1×10^{-4}
Weight decay (L_2)	1×10^{-4}
Dropout rate	0.4
LR scheduler patience	2 epochs
Early stopping patience	5 epochs
Optimizer	Adam
Loss function	Cross-entropy
Random seed	42

B. Training Dynamics

During training with the ResNet-18 transfer learning approach, the model achieved high training accuracy, approaching 100% in later epochs. However, validation accuracy remained in the range of approximately 40–50%, indicating significant overfitting. This behavior is consistent with the small size of the training dataset: the model has sufficient capacity to memorize the training examples but lacks sufficient diversity to learn generalizable features.

The learning rate scheduler activated during training, reducing the learning rate in response to stagnating validation performance. Early stopping was triggered before the maximum number of epochs was reached, preventing further overfitting.

C. Prediction Analysis

Table II presents sample prediction results from the trained model on previously unseen test images.

TABLE II
SAMPLE PREDICTION RESULTS

Test Image	Prediction	Confidence
Image 1	Digital	87.44%
Image 2	Traditional	58.03%

The results reveal that the model achieves high confidence on images with clear medium-specific characteristics (e.g., Image 1 with distinct digital properties) but produces lower confidence predictions for more ambiguous cases (e.g., Image 2). This pattern suggests that the model has learned some meaningful visual distinctions between the two classes but struggles with images where the two media exhibit overlapping characteristics, such as clean line art that could plausibly be either traditional or digital.

D. Error Analysis

Qualitative examination of misclassified and low-confidence predictions reveals several recurring failure modes:

1) **Clean traditional line art:** Traditional artwork with very clean, precise lines is sometimes misclassified as digital because it lacks the expected paper texture and stroke irregularity.
2) **Textured digital artwork:** Digital artwork that deliberately mimics traditional media textures (e.g., digital brush strokes simulating watercolor) confuses the model.
3) **High-quality scans:** Well-scanned traditional artwork with minimal scanning artifacts loses some of the physical media cues that the model relies upon.
4) **Dataset size effects:** With limited training examples, the model may overfit to incidental features (e.g., background color, image dimensions) rather than learning robust medium-specific representations.

V. DISCUSSION

A. Effectiveness of Transfer Learning

The transfer learning approach with ResNet-18 provides a more stable starting point than training from scratch, as the pretrained convolutional layers already encode useful low-level features such as edge detectors, texture analyzers, and color pattern recognizers. These features are directly relevant to the medium classification task, where the distinction between classes depends on textural and structural properties.

However, the benefits of transfer learning are limited by the small dataset size. While the frozen early layers provide useful generic features, the fine-tuned later layers and the classification head require sufficient data to specialize effectively. The observed overfitting indicates that the current dataset does not provide enough variation for the model to learn a robust decision boundary.

B. Impact of Dataset Size

The most significant limitation of this work is the small dataset size. Deep learning models typically require thousands to tens of thousands of labeled examples to generalize well. In this project, the dataset consists of a limited collection of original artwork, which constrains the diversity of visual patterns available for learning.

Data augmentation partially addresses this limitation by introducing geometric and photometric variability, but it cannot generate fundamentally new visual content. The augmentation strategy employed here (random crops, flips, rotations, and color jitter) is well-suited to the task, as these transformations preserve the medium-specific characteristics while altering global composition.

C. Architectural Considerations

The choice of ResNet-18 as the transfer learning backbone represents a balance between model capacity and the risk of overfitting. Larger architectures (e.g., ResNet-50, ResNet-101) offer greater representational power but also have significantly more parameters, which would exacerbate overfitting on a small dataset. The selective fine-tuning strategy, freezing all but the final residual block, further reduces the effective number of trainable parameters while allowing the model to adapt its higher-level representations to the target domain.

The dropout rate of 0.4 in the classification head provides additional regularization. This relatively aggressive dropout rate was chosen specifically to counteract the tendency toward overfitting observed during initial experiments.

VI. FUTURE WORK

Several directions for improvement are identified.

Dataset expansion. The most impactful improvement would be to substantially increase the dataset size and diversity. This could involve collecting artwork from multiple artists, including a wider range of traditional media (pencil, charcoal, watercolor, oil paint) and digital tools (vector graphics, 3D rendering).

Advanced augmentation. Techniques such as CutMix, MixUp, or style-based augmentation could introduce additional training variability without requiring new data collection.

Full fine-tuning. With a larger dataset, fine-tuning all layers of the pretrained network (with differential learning rates) could improve feature adaptation to the artwork domain.

Explainability analysis. Applying gradient-weighted class activation mapping (Grad-CAM) [10] to visualize which image regions most strongly influence model predictions would provide insight into the features the model has learned to associate with each medium.

Multi-class extension. Extending the classification beyond binary (traditional vs. digital) to include specific media categories (e.g., pen, pencil, watercolor, digital painting, vector art) would increase the practical utility of the system.

Ensemble methods. Combining predictions from multiple models (e.g., the baseline CNN and the fine-tuned ResNet) could improve robustness and reduce variance.

VII. Conclusion

This paper presented InkOrPixel, a CNN-based system for classifying artwork by creation medium. The system distinguishes between traditional artwork produced on physical media and digital artwork created with illustration software. Two approaches were investigated: a baseline CNN trained from scratch and a transfer learning approach using ResNet-18 with selective fine-tuning.

The results demonstrate that convolutional neural networks can learn medium-specific visual representations, particularly when leveraging features pretrained on large-scale datasets. The transfer learning approach showed higher confidence on unambiguous examples, correctly identifying clear digital and traditional artworks. However, both approaches face limitations due to the small dataset size, resulting in overfitting and reduced generalization to ambiguous cases.

The primary takeaway from this work is that the problem of medium classification is tractable using standard CNN architectures and transfer learning, but requires sufficiently large and diverse training data to achieve robust performance. The modular pipeline developed in this project provides a foundation for future work with expanded datasets and more sophisticated modeling techniques.

The complete source code and documentation for InkOrPixel are publicly available at https://github.com/aditiputrevu/InkOrPixel.

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References

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2012, pp. 1097–1105.

[2] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. International Conference on Learning Representations (ICLR)*, 2015.

[3] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1–9.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[5] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks?," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2014, pp. 3320–3328.

[6] S. Kornblith, J. Shlens, and Q. V. Le, "Do better ImageNet models transfer better?," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 2661–2671.

[7] L. A. Gatys, A. S. Ecker, and M. Bethge, "Image style transfer using convolutional neural networks," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2414–2423.

[8] S. Karayev et al., "Recognizing image style," in *Proc. British Machine Vision Conference (BMVC)*, 2014.

[9] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. International Conference on Learning Representations (ICLR)*, 2015.

[10] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

Appendix

Listing 1 shows the ResNet-18 transfer learning model definition. All convolutional layers up to and including `layer3` are frozen; only `layer4` and the replacement classification head are updated during training.

```python
import torch.nn as nn
from torchvision import models

def get_model():
    weights = models.ResNet18_Weights.DEFAULT
    model   = models.resnet18(weights=weights)

    # Freeze all pretrained layers
    for param in model.parameters():
        param.requires_grad = False

    # Unfreeze the final residual block for fine-tuning
    for param in model.layer4.parameters():
        param.requires_grad = True

    # Replace the 1000-class head with a binary classifier
    in_features = model.fc.in_features
    model.fc = nn.Sequential(
        nn.Dropout(p=0.4),
        nn.Linear(in_features, 2)
    )
    return model
```

Listing 1. ResNet-18 transfer learning model definition (`model.py`).

Listing 2 shows the augmentation pipeline applied during training. Validation and test images receive only resizing and normalization to ensure deterministic evaluation.

```python
from torchvision import transforms

train_transform = transforms.Compose([
    transforms.Resize((256, 256)),
    transforms.RandomResizedCrop(224, scale=(0.8, 1.0)),
    transforms.RandomHorizontalFlip(),
    transforms.RandomRotation(degrees=15),
    transforms.ColorJitter(
        brightness=0.2,
        contrast=0.2,
        saturation=0.2,
        hue=0.05
    ),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],
        std =[0.229, 0.224, 0.225]
    ),
])

val_test_transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],
        std =[0.229, 0.224, 0.225]
    ),
])
```

Listing 2. Training and evaluation transforms (`dataset.py`).